QIJUN LIAO

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EDUCATION BACKGROUND

University of Science and Technology Beijing (USTB) *Sep. 2024 – Present*
M.E. in Vehicle Engineering, School of Mechanical Engineering *Beijing, China*
Cumulative GPA: **88.1/100** (Top 20% in Major)
Awards: Second-Class Graduate Scholarship (2025), Merit Graduate Student (2025).

University of Science and Technology Beijing (USTB) *Sep. 2020 – June 2024*
B.E. in Vehicle Engineering, School of Mechanical Engineering *Beijing, China*
Cumulative GPA: **87.1/100** (Top 20% in Major, Waived from admission test for M.E. due to academic excellence).
Awards: Merit Student (2022, 2023), Outstanding Graduates (2024).

PUBLICATIONS

[1] **Qijun Liao**, Zhaoxin Yu, Jue Yang*, **"Constraint-Enhanced Reinforcement Learning Based on Dynamic Decoupled Spherical Radial Squashing"**, Under review in *NeurIPS 2026*.
https://arxiv.org/abs/2605.04185

[2] **Qijun Liao***, Jue Yang, **"Response-Aware Risk-Constrained Control Barrier Function With Application to Vehicles"**, Under review in *Automatica*. *Corresponding Author.
https://arxiv.org/abs/2603.24598

[3] **Qijun Liao**, Jue Yang*, Yiting Kang, Xinxin Zhao, Yong Zhang, Mingan Zhao, **"Hybrid Energy-Aware Reward Shaping: A Unified Lightweight Physics-Guided Methodology for Policy Optimization"**, Accept with Minor Revision in *Neurocomputing*. *Corresponding Author. https://arxiv.org/abs/2603.11600

[4] **Qijun Liao**, Jue Yang*, Subhash Rakheja, Yiting Kang, Yumeng Yao, Yuming Yin, **"Wheel Dynamic Load Estimation Method Based on Gas Pressure of Hydro-pneumatic Suspension"**, Under review in *Vehicle System Dynamics*. *Corresponding Author. https://arxiv.org/abs/2603.14242

RESEARCH EXPERIENCE

DD-SRad: Decoupled Dynamic Spherical Radial Squashing *Nov. 2025 – May. 2026*
Senior Author | *Submitted to NeurIPS 2026* *Under Review*

- **Differential Action-Space Parameterization:** Extended SRad to differential action space, enforcing actuator slew-rate limits $|a_t^i - a_{t-1}^i| \le \delta^i$ directly within the policy parameterization layer.
- **Dimension-wise Effective Radius:** Designed per-dimension adaptive radii R_{eff}^i to remove global-radius coupling and preserve feasible regions under heterogeneous actuator limits.
- **Gradient-Stable Optimization:** Introduced latent norm regularization $\lambda_{\text{base}}\mathbb{E}[\|u\|^2]$ to mitigate Jacobian saturation near constraint boundaries, with λ_{base} as the sole algorithm-level hyperparameter.
- **Empirical Validation:** Benchmarked against constrained RL and post-hoc projection baselines under homogeneous and heterogeneous rate constraints; validated legged locomotion in Isaac Lab with realistic joint limits.

H-EARS: Unified Lightweight Physics-Guided Policy Optimization *May. 2025 – Oct. 2025*
Senior Author | *Submitted in Neurocomputing* *Accept with Minor Revision*

- **Functional Independence:** Proved that task objectives and energy-efficiency terms can be optimized independently without gradient conflict under an additive reward structure.
- **Energy-Guided Convergence:** Established that energy-based potentials provide denser guidance than sparse task rewards, accelerating policy optimization through Lyapunov-like structure.
- **Lightweight Physical Modeling:** Reduced modeling complexity to $O(n)$ through selective energy characterization, avoiding full analytical dynamics for complex mechanical systems.

R^{2}CBF: Response-Aware Risk-Constrained CBF *Sep. 2025 – Jan. 2026*
Senior/Corresponding Author | *Submitted to Automatica* *Under Review*

- **Response-Aware Uncertainty Modeling:** Built stochastic response distributions from measurable vehicle body signals while preserving nominal dynamics for control-gradient structure.
- **CVaR-Based Safety Reformulation:** Reformulated deterministic CBF constraints into tail-risk constraints over barrier derivative distributions, replacing conservative worst-case bounds with CVaR margins.

- **Bayesian Online Adaptation:** Updated uncertainty covariance online via inverse-Wishart Bayesian inference driven by real-time prediction residuals.
- **Safety Validation:** Achieved zero safety-boundary violations across extreme combined-condition TruckSim scenarios, with empirical violation rates consistent with theoretical risk bounds.

Wheel Dynamic Load Estimation Method Based on Gas Pressure *Jan. 2025 – May. 2025*
Senior Author | *Submitted to Vehicle System Dynamics* *Under Review*

- **Fluid Inertia-Incorporated Modeling:** Developed a nonlinear gas-oil coupling model incorporating fluid inertia, polytropic gas dynamics, and flow-dependent damping.
- **Pressure-to-Load Observability:** Proved bijectivity and differentiability of the mapping $p \to v \to F_z$, enabling unique wheel-load reconstruction from pressure trajectories.
- **Experimental Validation:** Achieved force RMSE $<4.2\%$, velocity RMSE $<3.8\%$, and displacement RMSE $<4.9\%$ in bench tests; maintained wheel-load RMSE $<5\%$ in TruckSim co-simulation.

WORK EXPERIENCE

BAIC BJEV (Beijing Electric Vehicle) *Sep. 2025 – Present*
Supply Quality Engineer Intern | *Chassis Engineering Department, Digital Intelligence Group* *Beijing, China*
Developed RAG-based knowledge assistants and task-specific LLM skills for departmental regulations, audit procedures, and engineering workflows, deploying them on the company LLM platform to improve process retrieval and work efficiency.

Supported SQE tasks for the Vehicle Motion Domain Controller, including supplier quality review, issue tracking, and manufacturing-deviation analysis affecting chassis control performance.

SAIC-GM-Wuling Automobile (SGMW) *July 2023 – Aug. 2023*
R&D Intern | *Prototype Testing Department* *Liuzhou, China*
Performed technical review of DHT hybrid powertrain architecture and energy management strategies, synthesizing topology design, mode-switching logic, and control objectives into system-level engineering documentation. Supported prototype assembly and debugging, linking control-oriented design assumptions with practical constraints in packaging, manufacturing, and vehicle-level testing.

HONORS

3rd Prize, National, 14th National Zhou Peiyuan Mechanics Competition (**Highest in USTB History**) *2023*
2nd Prize, National, 18th National University Smart Car Competition (National Finals & Northern Division) *2023*
2nd Prize, Provincial, National University Physics Competition (**Rank 1st in School of Mechanical Engineering**) *2021*
1st Prize, University, USTB Robotics Competition *2021*

PROFESSIONAL SKILLS

Theory: Reinforcement Learning, Control Barrier Functions, Lyapunov Stability, Convex & Distributionally Robust Optimization, Sim-to-Real Transfer.
Programming: Python, PyTorch, TensorFlow, MATLAB/Simulink, C/C++, LaTeX
Simulation: Isaac Lab, TruckSim, CarSim, CARLA

CAMPUS SERVICE

Head of Academic Department, Student Physics Association, USTB *May 2021 – Present*
Designed and delivered training curriculum for the National Zhou Peiyuan Mechanics Competition; conducted specialized lectures on Theoretical Mechanics and dynamics. Organized weekly physics tutoring seminars for undergraduates.